Backtesting Value-at-Risk Models Under SR 11-7: A Comparative Analysis of Kupiec, Christoffersen, and Basel Traffic-Light Tests Applied to S&P 500 Returns (2018–2024)

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July 2026

ABSTRACT

This paper backtests three Value-at-Risk (VaR) models—Historical Simulation, Parametric (Normal), and Student-t—hold up when tested against S&P 500 daily returns covering January 2018 through December 2024. That seven-year run features four distinct market backdrops: the tail end of pre-COVID growth, the 2020 COVID-19 plunge and quick recovery, the bear market sparked by Fed rate hikes in 2022, and the rebound of 2023–2024. Lots of classic backtesting work just calibrates a VaR model once over a fixed period and then checks how it did, but that's not the approach here. Instead, this study uses a walk-forward design. Each model is recalibrated every day with the previous 250 trading days—and then checked against the next day's actual return. That gives us 1,509 out-of-sample forecast days between January 2019 and December 2024.

To judge the models, three big backtesting tests get used: Kupiec's Proportion of Failures test (1995), Christoffersen's Conditional Coverage test (1998), and the Basel Committee Traffic Light test. All these fits under the SR 11-7 model-risk-management framework, which was the rule of the land until the Fed's April 2026 switch to SR 26-2. The results aren't just a clear "winner" vs "loser"—things are more nuanced. At the tough 99% confidence level, all three models formally miss unconditional coverage, but the gap is wide across them. The Parametric Normal model is the worst offender by far (44 violations, LR = 36.92, $p < 0.0001$), while Historical Simulation (26 violations; LR = 6.55) and Student-t (28 violations; LR = 8.91) get much closer to what you'd want. But it doesn't stop there—violation clustering seems everywhere. All models fail the Christoffersen independence test at 95%, and only Student-t barely manages independence at 99% ($p = 0.107$). Its conditional-coverage test still rejects, though, thanks to coverage shortfall. Breaking down subperiods, the grinding 2022 rate-hiking regime shakes up violation clustering more than the sharper COVID shock did, implying these gradual shifts are harder for fixed-window estimators to follow than abrupt ones. On the final 250 days, none make it into Basel's Green zone anymore. The takeaway? Clustering during regime shifts is baked into fixed-window VaR estimation—it's not something you fix just by ditching the Normal distribution. This matters for SR 11-7 model validation, and, after SR 11-7's replacement with SR 26-2, for how the new rules put more emphasis on materiality in model supervision.

Keywords: Value-at-Risk, Backtesting, Kupiec Test, Christoffersen Test, Basel Traffic Light, SR 11-7, SR 26-2, Model Validation, Walk-Forward Backtesting, Historical Simulation, Student-t Distribution, Fat Tails, Market Risk

JEL Classification: G10, G17, G28, C52, C58

1. Introduction

Value-at-Risk (VaR) has been at the heart of financial risk management ever since J.P. Morgan introduced RiskMetrics back in the early 1990s. The idea's simple: VaR tells you the max expected loss over a certain period, at a set confidence level. That made it the go-to for regulatory capital under Basel II, Basel III, and the Fundamental Review of the Trading Book (FRTB). In the U.S., the Federal Reserve's Supervisory Guidance on Model Risk Management—SR 11-7 (2011)—put strict requirements on every quantitative model used for these purposes, VaR models included: independent validation, ongoing monitoring, and regular backtesting. SR 11-7 shaped model risk rules for fifteen years, until April 2026, when the Fed, OCC, and FDIC swapped it out for SR 26-2. We'll come back to that change later. For this study, since the 2018–2024 sample fits squarely within SR 11-7's reign, that's the lens we use, though we'll talk about what SR 26-2 changes.

VaR's big flaw: it leans on how returns are distributed, and the usual assumption—that returns are Normally distributed—is about as thoroughly debunked as anything gets in finance. Equity return series show fat tails, negative skew, and volatility clustering—especially during market stress, exactly when risk measures need to be right. The upshot: a Gaussian VaR model misses tail risk, just when banks need to see it most.

The 2018–2024 stretch in this study isn't an easy ride for VaR models. It dips into a late-2018 drawdown, the COVID crash in March 2020 (one of the fastest bear markets ever), and its quick rebound, the slow-burn Fed hiking bear market in 2022, and the 2023–2024 comeback. Rather than treating all seven years as a blob, this paper backtests each model out-of-sample, day by day, and breaks down the violations by market regime—which, it turns out, matters a lot.

This paper stands apart from most backtesting studies in a specific way. Most, including Halilbegovic et al. (2019) and Fruzzetti, Nasti, and Orlandi (2023)—the two we compare against—set up the VaR model once (maybe a few times) and measure violations against that static threshold. It's easy but blurs the line between fitting and actual forecasting. Here, every forecast for every model gets refit daily, using only what's known before that day. Each model's recalibrated on a trailing 250-day window and compared to the next day's actual return, a fresh run for all 1,509 out-of-sample days. That puts a lot more pressure on the models and changes what you see compared to a single static calibration.

There are four findings here that stand out. First, with all three models held to the same walk-forward standard, unconditional coverage failures at 99% aren't just a Gaussian thing—they show up across all three, though the Parametric Normal model is much worse (five to six times by LR stats) than Historical Simulation or Student-t. Second, clustering of violations is nearly universal too: every model fails Christoffersen's independence at 95%, and only Student-t squeaks by at 99%, so fat-tailed assumptions help with clustering much more than with coverage itself. Third, breaking out violations by regime shows the 2022 rate-hiking bear market triggered a higher proportional excess of violations than the COVID crash, which is surprising since COVID was a sharper shock. But apparently, a fixed 250-day window catches up to sudden shocks better than those slow, grinding changes that take months to reveal themselves—most of the year, really. Fourth, if you focus just on the most recent 250 trading days in 2024, none of these model's land in Basel's Green zone anymore, which matters right now for any firm leaning on these approaches. Altogether, this says violation clustering during regime shifts is baked into fixed trailing-window VaR estimation. You don't solve it just by swapping out Normal for Student-t. That drives work on regime-aware, machine-learning-based VaR models (see Section 7.2).

The rest of the paper is laid out like this:

Section 2 reviews the literature, showing how this study's walk-forward design fits in with recent work. Section 3 covers the data and basic stats. Section 4 explains each VaR estimation method and how the walk-forward works. Section 5 describes the backtesting tests. Section 6 dives into the results. Section 7 explores what all this means for SR 11-7 validation in practice—plus the April 2026 shift to SR 26-2—and what it suggests for building AI-powered VaR models. Section 8 wraps up and hints at next steps for research.

2. Literature Review

2.1 The Development of VaR as a Risk Measure

VaR landed in mainstream finance with J.P. Morgan's RiskMetrics (1994), then got locked in under Basel II (1996), which forced banks to hold capital against a 10-day 99% VaR estimate. That made VaR a real regulatory tool, not just an internal backstop, and suddenly getting the model right mattered. Jorion (2006) is still the go-to text for VaR; Christoffersen (2012) covers econometrics and the backtesting routines this paper uses.

VaR's flaws popped up almost as fast as it caught on. Decades before VaR was regulatory, Mandelbrot (1963) showed financial returns have fat tails, nothing like the Normal distribution—and this has been rediscovered so many times, it's a stylized fact. Artzner et al. (1999) took a different tack: VaR flunks the sub-additivity property (diversifying can, oddly, raise your VaR), and they pitched Expected Shortfall (ES), which fixes that problem. Basel III FRTB (2019) moved toward ES for market-risk capital, but VaR backtesting still sits at the heart of model validation (see Fruzzetti et al., 2023).

2.2 Backtesting Methodologies

Backtesting means you match up predicted losses (VaR forecasts) against what actually happened (realized losses). Zhang and Nadarajah (2017) count nearly thirty distinct backtesting techniques, split into four camps: unconditional coverage, conditional (independence), duration-based tests, and odds-and-ends like density forecasts. Kupiec (1995) comes up with the most popular unconditional coverage test, the Proportion of Failures (POF), which checks if violation frequency matches the stated confidence level, using a likelihood ratio. But as Zhang and Nadarajah (2017) and Haas (2001) point out, this test ignores when the violations happen—a model could nail the number but clump them all in one panic week!

Christoffersen (1998) closes that gap, creating the Conditional Coverage test that checks both unconditional coverage and independence—because clustering (losses on consecutive days, both over VaR) points to a model that's not keeping up with changing volatility. That's a bigger issue than just the average violation frequency, especially in a crisis. In Section 6, that split—coverage vs independence—shows up in the results: some models have decent coverage but still fall flat on independence, exactly what the Christoffersen test spots.

The Basel Committee's Traffic Light scheme (1996; 2019 revision) adds a blunt, but regulatory-heavy, layer: it sorts models into green, yellow, or red zones based simply on violation counts (over 250 days at 99% confidence), with the Yellow and Red zones jacking up capital requirements or triggering mandatory reviews. Since it's just the count, independence doesn't matter, so you can pass Basel but fail Christoffersen—or vice versa.

2.3 Fat Tails and Non-Normality in Financial Returns

Empirically, the Gaussian assumption for returns is dead. Fama (1965) showed daily stock returns are leptokurtic—fat tails—one of the earliest proofs of the point, and it's been confirmed everywhere since. McNeil, Frey, and Embrechts (2015) find excess kurtosis in the 5–15 range for daily equity returns, which matches the S&P 500's full-sample kurtosis of 14.41 in our study (see Section 3)—high but not shocking given the inclusion of the March 2020 crash.

The student-t distribution, with its freely adjustable tail thickness, is the main alternative to Normal in risk modeling; Blattberg and Gonedes (1974) suggested it for stocks and it's been standard since. Low degrees of freedom make for fatter tails; typical daily equity estimates run 3–7 degrees. Historical Simulation skips distributional assumptions entirely and uses recent realized returns—great for matching whatever happened in the calibration window, but dependent on those specifics.

2.4 SR 11-7, SR 26-2, and Model Risk Management

SR 11-7 (2011) steered model risk at U.S. financial institutions for fifteen years, defining models as "quantitative methods, systems, or approaches" that turn inputs into numbers using statistical/economic assumptions. VaR models fit the definition, so they had to undergo independent validation, monitoring, and backtesting—explicitly tying backtest results to model tweaks and restrictions.

April 17, 2026, brought the switch: SR 11-7 was replaced by SR 26-2 (Fed, OCC, FDIC, 2026), marking the first big revision since 2011. SR 26-2 keeps the core principles—independent challenge, validation, monitoring—but shifts the focus toward model materiality: tailoring expectations by risk profile, instead of "one-size-fits-all," and narrows the "model" definition (leaving out basic deterministic calculations SR 11-7 had swept in). For this paper, since the sample is pure SR 11-7 era and the backtesting routines are still core disciplines under SR 26-2, we stick with SR 11-7 as the benchmark and later look at how SR 26-2 changes things.

2.5 Positioning This Study

Two recent papers come close to this one in scope. Halilbegovic, Celebic, Arapovic, and Vehabovic (2019) backtest Historical Simulation, Parametric, and Monte Carlo VaR against a five-stock portfolio, using five backtests (POF, TUFF, Basel, Christoffersen, Mixed Kupiec), over just one calm year and calibrating from the static full sample. Fruzzetti, Nasti, and Orlandi (2023), with Bank of Italy data (2012–2021), backtest RiskMetrics VaR and expand into ES backtesting, using rolling windows for robustness but not as the main analysis.

This study diverges from theirs in three big ways. First, scope: instead of just a handful of stocks or currency portfolios, we test a broad equity benchmark, traversing seven turbulent years and several regimes—the only way we could break out the violation patterns in Section 6.5. Second, estimation: where Halilbegovic et al. calibrate once and Fruzzetti et al. use rolling windows mainly as robustness checks, we deploy daily walk-forward re-estimation, so every forecast uses only data available before that day. Third, interpretation: a calm one-year sample lets Halilbegovic et al. declare their models "accurate at almost all levels," while Fruzzetti et al. see their RiskMetrics VaR pass standard tests and ES fail. Our walk-forward, multi-regime approach, in contrast, finds coverage failures and clusterings everywhere, across all models—the differences are just degree, not pass/fail. That's evidence that conclusions about VaR adequacy depend a lot on the sample's market regimes and the estimation design behind the forecasts, something that, frankly, deserves more attention in this space.

3. Data and Descriptive Statistics

3.1 Data Description

I'm working with S&P 500 daily closing prices, covering January 3, 2018, through December 30, 2024—1,759 trading days in total, all pulled from Yahoo Finance. The S&P 500 makes sense as a test case. It's not just the go-to U.S. equity benchmark; regulatory guidance (SR 11-7, now SR 26-2) centers on it, and its run from 2018 to 2024 includes just about every market mood: low-volatility stretches, a global crisis, a raging rally, and a stubborn bear market. The mix makes for a tough but realistic proving ground for Value at Risk (VaR) models—much better than a smooth or shorter history.

For returns, I calculated daily log returns by taking the natural log of the ratio between consecutive closing prices. Table 1 sums up the stats for both the entire period and five natural "regimes": pre-COVID, the COVID crash, the post-crisis bull run, rate-hike bear market, and the more recent recovery.

Table 1 — S&P 500 Descriptive Statistics, 2018–2024

Period	Obs.	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Excess Kurtosis
Full Sample (2018–2024)	1,759	0.045	1.248	−12.765	8.968	−0.816	14.415
Pre-COVID (2018–2019)	502	0.036	0.944	−4.184	4.840	−0.612	3.641
COVID Crisis (2020)	253	0.060	2.185	−12.765	8.968	−0.866	8.507
Post-COVID Bull (2021)	252	0.095	0.826	−2.601	2.351	−0.369	0.698
Rate Hike Bear (2022)	251	−0.086	1.524	−4.420	5.395	−0.009	0.336
Recovery (2023–2024)	501	0.086	0.811	−3.043	2.498	−0.285	0.734

Source: Author calculations from S&P 500 daily closing levels (Yahoo Finance). Kurtosis figures are excess kurtosis (Normal = 0). Full-sample and subperiod figures use calendar-year boundaries; Section 3.2 and Table 1 describe the full estimation sample, which differs from the out-of-sample walk-forward test window used in Tables 3–7 and defined in Section 4.

3.2 Distributional Properties

If you look at Table 1, three things jump out. First, the full sample's excess kurtosis—14.41—is extremely high. That basically screams "fat tails." McNeil et al. (2015) list 14 or so as about the wildest, you'll ever see in daily equity returns, which suggests the ordinary Gaussian VaR model is bound to underestimate tail risk here.

Second, the Jarque–Bera test gives a statistic of 15,424.4 ($p < 0.0001$). There's no arguing with that; this data isn't close to normal. And third, the full-sample skewness is clearly negative at −0.816. That means the left tail—the side with the big losses—is a lot longer than the right. Despite a common belief that returns skew positive, equities usually sell off harder than they rally. This lopsidedness (often called the "leverage effect") is well-documented and reflects those nasty crash days. Any model that bets on symmetric or positively skewed returns here is simply ignoring the risk we care about. Breaking things down by regime, you see these quirks aren't always present. For example, during the worst of the COVID crash (2020), volatility peaks at 2.19% and skewness bottoms out at −0.87. That's exactly what you expect from a stretch with steep down days and a rapid snapback. The 2022 "rate hike" bear market looks different—it's choppier but not as extreme (volatility = 1.52%), practically no skewness (−0.01), and almost normal kurtosis (0.34). Instead of sudden plunges, markets just steadily bled lower. This divide—between quick, scary drops versus long, grinding slides—comes up again later. In Section 6.5, I explain why this matters a lot for fixed-window VaR models, sometimes more than volatility itself.

4. VaR Estimation Methodologies

Each of the three VaR models is tested out-of-sample in a walk-forward fashion. Here's the process: I start calculating VaR predictions once I have 250 past daily returns, which means real testing starts on January 2, 2019, and runs to December 30, 2024—1,509 trading days. Every day, each model trains on just the past 250 returns and spits out a VaR forecast for the next day—

always staying "blind" to future data. This way, any differences between models reflect their design, not the amount or type of data they use.

4.1 Historical Simulation

Historical Simulation (HS) is still king for banks and risk managers—mostly because it doesn't assume anything about returns' behavior and is easy to explain. At its core, the HS VaR for day t at confidence level α is just the empirical (1−α)-th percentile of the last 250 returns:

$$VaR_t(\alpha) = -Q_{1-\alpha}(r_{t-250}, \ldots, r_{t-1})$$

Here, Q is the empirical quantile function. With HS, whatever "weirdness" is in recent data—fat tails, skew, clustering—is captured automatically. That's both its strength and its Achilles' heel: if the market regime shifts but hasn't fully filtered into the last 250 days, HS will almost always understate risk during that period.

4.2 Parametric (Normal) VaR

This is the textbook approach—sometimes called "Variance-Covariance VaR." It treats the past 250 days as normally distributed, estimating the mean and standard deviation from just that window:

$$VaR_t(\alpha) = -(\mu_t + z_{1-\alpha} \cdot \sigma_t)$$

$z_{1-\alpha}$ is the (1−α) quantile of a standard normal ($z = -1.645$ for 95%, −2.326 for 99%). I re-estimate μ and σ every day on the same rolling window as HS to keep things fair. Any differences in performance, then, trace back to the assumption of normality—not different datasets. The Normal VaR is fast and simple, but Section 3 makes it clear: normality is a pretty bad fit here.

4.3 Student-t VaR

The student-t model tries to address "fat tails" directly. It adds a degrees-of-freedom parameter, ν, which tunes how thick the tails are:

$$VaR_t(\alpha) = -(\mu_t + t_{1-\alpha,\nu} \cdot \sigma_t)$$

Here, t1−α, ν is the relevant quantile of the standard Student-t (which becomes normal as $\nu \to \infty$). I estimate μ, σ, and ν by maximum likelihood from the most recent 250 returns—refitting daily, like the other models. Because ν is estimated on just these 250 returns, its day-to-day value bounces around more than if I fit one value to the whole sample. But that's the cost of keeping things truly out-of-sample. In Section 6.1, I show how this approach shifts estimated VaR over time.

5. Backtesting Framework

5.1 Kupiec Proportion of Failures Test

Here's how the Kupiec Proportion of Failures test works. For each day, you set an indicator I(t): it's 1 if the return that day falls below the negative value-at-risk (VaR) forecast—a VaR "break"—and 0 if not. Over T days out of sample, you count up the total violations: $N = \Sigma I(t)$. The sample violation rate is just $\hat{p} = N/T$.

Kupiec (1995) tests whether this observed rate lines up with the model's expected failure rate, under the null hypothesis H_0: $\hat{p} = 1-\alpha$ (so if your VaR is at 99%, you expect about 1% violations). The likelihood-ratio statistics are:

$$LR^{POF} = -2 \ln[(1-p)^{T-N} \cdot p^N] + 2 \ln[(1-\hat{p})^{T-N} \cdot \hat{p}^N]$$

This statistic follows a chi-squared distribution with one degree of freedom if H_0 holds. If the actual violation rate is too high or too low compared to what the model forecasts, that's a problem—the test rejects the model. If you're being overly cautious (too few breaks), you're holding more capital than you need. If your model underestimates risk (too many breaks), you're not holding enough capital—and that's a serious issue. Regulators (under SR 11-7) expect banks to document these failures and lay out a plan to fix them if the model flunks the Kupiec test repeatedly at a validated confidence level.

5.2 Christoffersen Conditional Coverage Test

Christoffersen (1998) takes the Kupiec test further. Now we don't just ask, "Is the violation rate right on average?" We also check: "Do violations show up randomly, or in bursts?"

To do this, you note how often you see a violation on day t+1, depending on what happened on day t. You define $\pi(0,1)$ as the probability of a violation tomorrow when today saw none, and $\pi(1,1)$ as the probability of a violation tomorrow, given today already had one.

The likelihood-ratio test is:

$$LR^{ind} = -2 \ln[L(\hat{\pi})] + 2 \ln[L(\hat{\pi}_{01}, \hat{\pi}_{11})]$$

The first term is the likelihood of assuming independence (violations don't depend on yesterday), and the second is for the unrestricted Markov chain (violations can cluster). LRind also follows a chi-squared distribution with one degree of freedom.

To check both coverage and independence together, you simply add up the Kupiec and independence statistics—now you've got a chi-squared test with two degrees of freedom. Sometimes, models can hit the right average rate but still fail the independence part—meaning violations arrive in clusters rather than being spread out. That's not uncommon—Section 6.3 finds clustered breaks are the rule, not the exception, in practice.

5.3 Basel Traffic Light Test

Regulators also use the Basel Traffic Light test, which is a bit more straightforward. You look back over the past 250 trading days at the 99% VaR and count how many violations pop up. Based on this count, you fall into a green, yellow, or red zone, which triggers different capital multipliers and regulatory responses:

Table 2 — Basel Traffic Light Test: Zone Classification

Zone	Violations (250-day)	Capital Multiplier	Regulatory Response
Green	0–4	3.00×	No action required
Yellow	5	3.40×	Increased capital charge
Yellow	6	3.50×	Increased capital charge
Yellow	7	3.65×	Increased capital charge
Yellow	8	3.75×	Increased capital charge
Yellow	9	3.85×	Increased capital charge
Red	10 or more	4.00×	Mandatory model review; regulatory scrutiny

Source: Basel Committee on Banking Supervision (2019), "Supervisory Framework for the Use of Backtesting in Conjunction with the Internal Models Approach to Market Risk Capital Requirements.

6. Empirical Results

6.1 VaR Estimates

Table 3 pulls together the daily VaR forecasts for each model over the 1,509-day out-of-sample walk-forward window. Since each day gets a fresh estimate, the models spit out a whole range of VaR numbers, not just a single value. That spread is informative all by itself, it shows how quick each model is to react when the market shifts inside its trailing window.

Table 3 — VaR Estimates, Walk-Forward Out-of-Sample Window (Jan. 2019–Dec. 2024)

Method	Confidence	Mean VaR (%)	Min VaR (%)	Max VaR (%)	Final VaR (%)
Historical Simulation	95%	1.996	1.031	3.419	1.355
Parametric (Normal)	95%	1.941	1.060	3.577	1.221
Student-t	95%	1.718	0.996	3.051	1.131
Historical Simulation	99%	3.410	1.482	7.006	2.242
Parametric (Normal)	99%	2.764	1.538	5.081	1.764
Student-t	99%	3.466	1.554	9.159	1.999

Note: "Final VaR" is the forecast produced for December 30, 2024, the last day of the sample. Mean ratio of 99%/95% VaR: Historical Simulation 1.71×, Parametric 1.42×, Student-t 2.02×, against a Normal-distribution benchmark ratio of $z_{0.99}/z_{0.95} \approx 1.414$*. Deviations above the benchmark indicate fatter-than-Normal tails.*

Now, the ratio of mean 99% to mean 95% VaR gives a quick sense-check. If your returns are truly Normal, that ratio should be about 1.414, matching the z-scores. The Parametric Normal does this nearly perfectly (1.42×), because it's baked right into the model. But Historical Simulation (1.71×) and especially Student-t (2.02×) both overshoot that mark. And that's exactly what you'd expect with fat tails—these models see more risk in the extremes, which lines up with the non-Normality seen back in Table 1.

6.2 Kupiec Test Results

Table 4 lays out the Kupiec POF test results for all six model-confidence combos over that same 1,509-day window.

Table 4 — Kupiec Proportion of Failures Test Results (Out-of-Sample, T = 1,509)

Model	Conf.	Violations (N)	Expected	$\hat{p}$ (%)	LR Statistic	p-value	Decision
Historical Simulation	95%	71	75.5	4.71	0.282	0.596	PASS
Parametric (Normal)	95%	78	75.5	5.17	0.090	0.764	PASS
Student-t	95%	88	75.5	5.83	2.091	0.148	PASS
Historical Simulation	99%	26	15.1	1.72	6.551	0.010	REJECT*
Parametric (Normal)	99%	44	15.1	2.92	36.917	<0.0001	REJECT***
Student-t	99%	28	15.1	1.86	8.910	0.003	REJECT**

*Note: *** $p<0.001$; ** $p<0.01$; * $p<0.05$. H_0: observed violation rate equals theoretical rate ($\hat{p} = 1-\alpha$). LR statistic follows $\chi^2(1)$ under H_0.*

At the 95% level, all three models pass. The observed violation rates—4.71%, 5.17%, and 5.83%—land right around the expected 5%. None of the LR statistics are even close to grounds for concern. But things change fast at the 99% confidence level; now, all models fail the test, though not equally.

Parametric Normal stands out for all the wrong reasons: 44 violations when the expectation was just 15.1, so nearly triple the target rate, and its LR statistic absolutely dwarfs the others (36.9, compared to 6.6 and 8.9). Historical Simulation and Student-t- aren't perfect, but they're much closer to the target (26 and 28 violations, or 1.72× and 1.86× theoretical). In plain terms: Historical Simulation and Student-t almost get there. Parametric Normal misses by a mile—exactly as you'd expect based on what most literature says about Gaussian VaR underestimating tail risk (see Section 2.3). Still, none of the models are hitting the mark at this high a threshold.

6.3 Christoffersen Conditional Coverage Test

Table 5 shifts to the Christoffersen independence and joint conditional-coverage checks—a more nuanced look at whether violations follow each other around. And here's where things get interesting: violation clustering happens across all models. It's not a quirk of any one approach.

Table 5 — Christoffersen Conditional Coverage Test Results (Out-of-Sample)

Model	Conf.	π_{01} (%)	π_{11} (%)	LR Ind. (p)	LR CC (p)	Ind.	Cond. Cov.
Historical Simulation	95%	4.24	14.08	9.989 (0.002)	10.270 (0.006)	REJECT	REJECT
Parametric (Normal)	95%	4.69	14.10	9.526 (0.002)	9.616 (0.008)	REJECT	REJECT
Student-t	95%	5.28	14.77	10.008 (0.002)	12.099 (0.002)	REJECT	REJECT
Historical Simulation	99%	1.55	11.54	6.834 (0.009)	13.386 (0.001)	REJECT	REJECT
Parametric (Normal)	99%	2.73	9.09	4.017 (0.045)	40.934 (<0.0001)	REJECT	REJECT
Student-t	99%	1.76	7.14	2.593 (0.107)	11.502 (0.003)	PASS	REJECT

Note: π_{01} = probability of a violation tomorrow given no violation today; π_{11} = probability of a violation tomorrow given a violation today. LR Ind. follows $\chi^2(1)$; LR CC follows $\chi^2(2)$. Rejection at the 5% significance level.

At 95% confidence, every single model fails both the independence and joint tests. The chance of a violation following another violation (π_{11}) is about three times higher than the base chance after a quiet day (π_{01}). For example, with the Student-t, a violation today means it's 14.8% likely you'll see another tomorrow, versus just 5.3% after a calm day. That's a big difference—and a clear sign these models tend to bunch up their misses, even if the average number of violations looks fine on paper (see the Kupiec results above).

At 99%, the breakdown is more revealing. Historical Simulation and Parametric Normal both still bunch up their misses—failures all around. Only the Student-t model passes the independence test at this level ($p = 0.107$), hinting that its flexibility pays off as things get extreme. But it still fails the joint test, because, as shown earlier, it's still falling short on total coverage—the point of Christoffersen's joint test is that a model can't slip by just by getting one part right.

6.4 Basel Traffic Light Test

Table 6 switches gears, reporting Basel Traffic Light results for just the most recent 250 trading days (all of 2024), with a 99% confidence bar—this is how regulators would size up a model.

Table 6 — Basel Traffic Light Test Results (Trailing 250 Days, 99% CI, Jan.–Dec. 2024)

Model	Violations	Expected	Zone	Capital Multiplier	Status
Historical Simulation	6	2.5	Yellow	3.50×	Increased charge
Parametric (Normal)	6	2.5	Yellow	3.50×	Increased charge
Student-t	5	2.5	Yellow	3.40×	Increased charge

Note: 250-day evaluation window runs January 3, 2024, through December 30, 2024. Regime-specific results for the 2020 and 2022 windows are reported separately in Table 7.

None of the models scores a Green here. All fall into the yellow zone for 2024, with Historical Simulation and Parametric Normal showing six violations (vs. 2.5 expected) and Student-t-five.

This matters because, by volatility, 2024 was pretty tame for the S&P 500. In earlier studies using static, full-sample calibrations, these models probably would have stayed in the Green zone—less real pressure in the recent period. But the rolling, "what-have-you-done-for-me-lately" test here asks if recalibrating only on recent history kept up, even during a relatively mild year. For all three models here, the answer is no.

6.5 Subperiod Analysis

Now, Table 7 breaks things down by market regime. The out-of-sample test starts in 2019 (the initial 250-day window is used up by 2018). The results are classic during the COVID Crisis: violation rates pop way above what any model expected, especially for Parametric Normal (14 violations vs. 2.5 expected, up 453%), and to a lesser—but still big—extent for Historical Simulation and Student-t (8 each, or 216% excess).

Table 7 — Subperiod Analysis: 99% VaR Violations, Out-of-Sample

Period	Obs.	Expected	HS Viol. (Rate, Excess)	Param Viol. (Rate, Excess)	Student-t Viol. (Rate, Excess)
Pre-COVID (2019)	252	2.5	1 (0.4%, −60%)	4 (1.6%, +59%)	1 (0.4%, −60%)
COVID Crisis (2020)	253	2.5	8 (3.2%, +216%)	14 (5.5%, +453%)	8 (3.2%, +216%)
Post-COVID Bull (2021)	252	2.5	1 (0.4%, −60%)	3 (1.2%, +19%)	1 (0.4%, −60%)
Rate Hike Bear (2022)	251	2.5	10 (4.0%, +298%)	17 (6.8%, +577%)	13 (5.2%, +418%)
Recovery (2023–2024)	501	5.0	6 (1.2%, +20%)	6 (1.2%, +20%)	5 (1.0%, −0%)

Note: HS = Historical Simulation; Param = Parametric (Normal). Excess (%) = (Actual − Expected) / Expected × 100. Violation counts across all five subperiods sum exactly to the full out-of-sample totals in Table 4 (HS: 1+8+1+10+6=26; Param: 4+14+3+17+6=44; Student-t: 1+8+1+13+5=28).

Here's the real surprise: the 2022 Rate Hike Bear market produces even bigger proportional miss rates for all three models—up to 577% excess for Parametric, 418% for Student-t, and 298% for Historical Simulation. This is weird at first, since COVID seems, by almost any measure (see Table 1), like the more dramatic shock. But there's a catch: COVID hit fast and hard, so a 250-day window filled up quickly with high-volatility days, letting models adapt. The 2022 cycle was drawn-out; much of the 250-day window kept including low-volatility observations from the 2021 bull run. So, the models kept trailing behind, leaning on stale calm data, and struggled to get on top of the new, choppier regime. For 2021 (the bull run) and the 2023–2024 "recovery," all three models report violation rates at or below expectations, in line with genuinely quiet markets—not just models getting "lucky" because they were slow to react.

Overall, you see that these rolling, day-by-day VaR forecasts give a more demanding, honest read on model performance—and almost all the models, even the flexible ones, struggle to keep up with fast-moving or persistent regime changes. Static, retrospective tests just can't spot those blind spots.

7. SR 11-7 Implications and Practical Guidance

7.1 Model Validation Requirements Under SR 11-7 and SR 26-2

SR 11-7 tells validators to really dig into model performance, not just tick boxes. It wants a clear measure of how well a model works — and just as important, where it doesn't. The results from Section 6 go straight at that, and they do more than just hand down a simple pass/fail grade. You walk away with some specifics.

First, Parametric Normal VaR has some tough limitations at the 99% confidence level: its violation rate is almost three times what theory predicts, and its LR statistics are four to six times higher than the alternatives. It's also worse when the market's stressed, not just in normal periods. That has to be in the validation record.

But it's not as simple as just swapping Normal for Historical Simulation or Student-t. Both alternatives still fall short on unconditional coverage at 99% when you run a walk-forward backtest. And, if you look at the independence piece using the Christoffersen test, every model — even Student-t at 95%, fails somewhere. So, "swap out Normal and you're safe" misses a big part of the problem.

Third, the kind of subperiod backtesting in Table 7 should be standard practice. Section 6.5's main point — that the market's 2022 regime shift showed even worse relative clustering than the COVID crisis — shows how much you'd miss if you only ran an aggregate, full-sample test. This is exactly the sort of regime-specific failure SR 11-7 wants to see flagged.

Fourth, the Basel Traffic Light test (Table 6) says all three models land in the yellow zone for the most recent 250 days, even though it wasn't an especially wild year. For any bank still relying on one of these models — with no regime-detection or volatility overlay — that's a real concern right now, not just a historical footnote.

SR 26-2 still expects strong independent validation and ongoing checking — it just wants the amount of effort to scale to the model's materiality and the bank's risks. For a VaR model that feeds straight into capital, you can bet that still means thorough documentation and backtesting, just as before. Maybe the paperwork is more proportional for simpler or less important models now, but for core capital models, all the key findings still need to be front and center. The requirement to chase down and document violation clustering isn't going away.

7.2 Toward AI-Augmented VaR Systems

Section 6.5's finding is pointed: slow, ongoing regime changes are tougher on a fixed 250-day estimator than sudden shocks. All three models tested here use a fixed-length window, so they're a step behind when a regime shifts — there's nothing built in to notice a change until it's already almost done. The Korvane ML-LiqVaR framework (Neupane, forthcoming) is designed to plug that gap. It uses a GARCH-Markov Switching piece to spot changes early, an XGBoost layer to pick up on nonlinear links between macro variables and the actual swings you see, and an LSTM to capture how violations tend to bunch up. Validators can break down each forecast with SHAP explainability — so it's not a black box, which SR 11-7 and SR 26-2 both demands.

But more complexity doesn't get you off the hook for validation. If anything, the bar goes up. The standard three-test framework here — Kupiec, Christoffersen, Basel Traffic Light — under a real walk-forward design (not some "in-sample" trick) is just the starting point. Any new regime-aware model out of the Korvane program gets checked first for whether it closes the independence issue in Table 5, and second for whether it dulls the regime transition sensitivity in Table 7. Those are the real gaps, not just vague "not fat-tailed enough" complaints.

8. Conclusion

This study backtested three popular VaR methods — Historical Simulation, Parametric Normal, and Student-t — on S&P 500 daily returns from 2018 to 2024. The design was a genuine walk-forward, with three tests: Kupiec, Christoffersen, and Basel's Traffic Light. Here's what stands out:

1. All three models fail unconditional coverage at 99%, not just the Parametric Normal — though Normal's the worst, with 44 violations when 15 were expected (LR = 36.92, $p < 0.0001$). Historical Simulation had 26 (LR = 6.55), Student-t had 28 (LR = 8.91). At 95%, everyone passes. So, the trouble is at the extremes, where distributional assumptions matter most.

2. Violation clustering is everywhere. Every model fails Christoffersen independence at 95%; only Student-t passes at 99%, and even that's not enough, because its joint conditional coverage is still rejected. Passing one test doesn't mean you've passed the other. A static, full-sample test can easily hide these gaps.

3. When you break it up by subperiod, the slow burning 2022 bear market saw more extra violations, model for model, than the sharp COVID crash. That's surprising, given COVID was wilder by old-school metrics. The reason: fixed windows absorb shocks quickly, but chase regime shifts for much longer.

4. Looking at just the latest 250 days — 2024, which wasn't abnormally volatile — none of the models make it into Basel's Green zone. That's not an academic finding; it matters for any shop still using these specs with no overlay.

5. These findings collectively motivate a specific, rather than generic, direction for model improvement: the failure mode common to all three models is not merely "insufficiently fat-tailed" but structurally an inability to detect a regime transition before it has substantially entered a fixed estimation window. I take up that specific gap directly in the next paper of this series, which introduces ML-LiqVaR, a regime-aware, machine-learning-augmented VaR model; the walk-forward, three-test methodology established here is intended to serve as its validation baseline.

One last point: How your backtest matters. Static, full-sample VaR backtests can tell a very different story compared to walk-forward tests — even with the same data and tools. Most published work still goes the static route. That gap could be used more attention. Down the road, this research group will push walk-forward methods to Expected Shortfall, multi-asset portfolios, and ML-augmented, regime-aware VaR systems.

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